

ZIDANE VIRANI

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Education

Western University

Bachelor of Engineering in Software Engineering

Apr 2028

GPA: 3.8/4.0

- Courses: Data Structures & Algorithms, Operating Systems, Neural Networks, NLP, Cloud Computing, Distributed Systems, Database Systems, Computer Networks, Software Architecture, Parallel Computing, Computer Vision, Statistical Learning
- Involvements: Outreach chair of Western Dental Association, Director at Western AI, Academic Mentor for Freshmen

Skills

- Python, C/C++, Go, Swift, JavaScript, SQL, Bash, Node.js, React, FastAPI, Django, Redis, Hugging Face, LangChain, CI/CD
- CUDA, AWS, Azure, PyTorch, Tensorflow, Keras, Docker, Kubernetes, GCP, PostgreSQL, Kafka, ONNX, Triton, OpenCV, FAISS

Experience

Founder & Lead Engineer, AI Content Engine ([Blog Link](#))

Jan 2023 – Dec 2024

- Sold a Python Retrieval Augmented Generation (RAG) pipeline for \$50,000 that scraped trending topics and auto-generated SEO-optimized articles with GPT-4, scaling my blog to 120K+ monthly readers and \$2,800/month in net profit
- Developed and deployed an end-to-end production-ready ML ranking model that increased article engagement by 412% by training on 10,000+ user-labeled examples using PyTorch, CUDA, Hugging Face Transformers, and multi-GPU EC2 infrastructure
- Reduced infrastructure costs by 57% by fine-tuning and deploying open-source LLM for SEO workflows and deploying it on EC2 Spot, building an inference pipeline that successfully processed all workloads previously done by GPT-4

Bloomberg: Incoming Machine Learning Engineer Intern

Jun 2026 – Aug 2026

Shopify: Applied Machine Learning Engineer Intern

Jan 2026 - Apr 2026

- Built an autocomplete feature for Shopify Sidekick using Python, PyTorch, Hugging Face Transformers, and a fine-tuned Qwen-0.6B LLM, reducing merchant query time by 68% and deploying to 5 Million+ active stores through Docker and Kubernetes
- Optimized Sidekick's real-time, low-latency ML inference pipeline using Python, PyTorch, ONNX, and Redis caching, cutting average response latency by 85% to enable sub-second user interaction, reducing infrastructure costs by 34%
- Engineered an automated data curation pipeline using PyTorch, SageMaker, and Spark to filter and rank 2M+ merchant queries through distributional analysis, boosting fine-tuned Qwen model eval performance by 3.8× on internal benchmarks

Node App: Machine Learning Developer Intern

Sep 2025 – Jan 2026

- Built an AI-powered search system (RAG) handling 5M+ documents using Python, Elasticsearch, and Docker, reducing response times by 92% through vector indexing and optimized caching techniques
- Improved search relevance by 91% by training a PyTorch ranking model on 500K+ temporal user-behavior events, rigorously validating results through end-to-end A/B experiments with statistical significance testing, then deploying via FastAPI on AWS
- Scaled the platform to handle 3× more traffic by using Docker for model containers, ONNX Runtime for faster inference, and Kubernetes autoscaling, reducing memory usage by 58% with sub-200ms latency

Western University: Machine Learning Engineer Intern

Jun 2025 – Sep 2025

- Built a real-time multimodal ML pipeline in Python and PyTorch fusing time-series audio signals, video frames, and text features to detect interruption events with low-latency streaming inference, deploying to 70+ political science departments for live analysis
- Led a team of 5 (researchers and engineers) to build an LLM-powered fact-checking system in Python, Django, PyTorch, FAISS, Docker, and AWS EC2 Spot, delivering a retrieval-and-verification pipeline that reached 92% precision across 1.2M papers

Vector Institute: Full-Stack Machine Learning Developer Intern

Apr 2025 - Jun 2025

- Built and deployed a senior-care voice assistant on AWS with Python, TensorFlow, Whisper and FastAPI, rolling it out across 18 retirement facilities (5,000+ users) and saving staff 1,200+ hours monthly through automated requests and daily care workflows
- Enabled multilingual support by fine-tuning OpenAI Whisper with PyTorch and LoRA on 50K+ caregiver-patient samples, boosting transcription accuracy by 34% and cutting clarification calls by 41% across all facilities

Dibbly Inc: AI Engineer Intern

Jan 2025 - Apr 2025

- Built a LLM-Powered chatbot (RAG) replacing a 3-day manual workflow for 5,000+ users, resolving 3,000+ weekly queries at 95% accuracy by deploying semantic search over 10,000+ documentation pages using LangChain and Pinecone vector database
- Optimized Mistral-7B (Open-source Large Language Model) fine-tuning in Python with CUDA, PyTorch, and AWS SageMaker, making training 28% faster by using more efficient fine-tuning techniques like LoRA and gradient checkpointing

Leadership

Aga Khan Development Network: Chair

Feb 2026 – Present

- Collaborated with 6 senior engineers and Canadian healthcare leaders to architect an AI-powered women-empowerment platform in India using Python, PyTorch, FastAPI, React, and AWS, securing \$100,000 in funding and deploying ML-driven matching to connect users with fair-wage opportunities